

Design & Reference Manual: A Local, Free Options-Flow & Quant-Trading Application (US Equities/ETFs)

Prepared for: Pure (Kasetsart University, Faculty of Engineering) — strong engineer, little finance background. **Date:** 13 September 2026. **Market focus:** US equity & ETF options. **Deployment:** local, free-to-use.

How to read this manual: Every financial term is defined in plain English the first time it appears, followed by a formula and a worked number where relevant. This is an educational engineering reference, **not investment, legal, or tax advice**.

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PART 1 — ALGO TRADING, OPTION-BUYING SIGNALS, AND SIGNAL ANALYSIS

1.1 What is algorithmic / quantitative trading?

Algorithmic trading means using explicit, coded rules to decide what to trade, when, and how much. Quantitative ("quant") trading means those rules are derived from data and statistics rather than gut feeling. For you, "the app" is a decision-support and research system, not necessarily an auto-executor.

Strategy categories (and which are realistic for a solo local developer)

- **Trend following / momentum** — buy what's going up, sell what's going down. Realistic for retail; robust; works on daily data. **RECOMMENDED starting point.**
- **Mean reversion** — bet that stretched prices snap back. Realistic on liquid names; sensitive to transaction costs.
- **Statistical arbitrage / pairs** — trade relative mispricing between correlated instruments. Feasible but data-hungry.
- **Market making** — quote both sides, earn the spread. **NOT realistic** for retail (needs colocation, exchange rebates, ultra-low latency).
- **Event-driven** — earnings, M&A, index rebalances. Partly realistic (earnings flow), but news latency hurts retail.
- **Volatility arbitrage** — trade implied vs realized volatility. Realistic conceptually (options), but needs good vol data and hedging discipline.
- **Options-flow "follow the smart money"** — the theme of the reference site. Realistic as a *signal-generation/context* layer; see honest caveats in §1.6.

Bottom line: as a solo developer on free/low-cost data with 15-minute delayed feeds, focus on daily/swing-horizon momentum, mean-reversion, and options-flow-derived *context* signals. Avoid anything latency-sensitive.

1.2 The full quant pipeline

Data ingestion → Cleaning/validation → Feature engineering → Signal generation
→ Filtering/ranking → Position sizing → Risk management → Execution → Post-trade analytics

- **Ingestion:** pull option chains, quotes, OHLCV on a schedule.
- **Cleaning:** handle missing rows, bad ticks, corporate actions, expired contracts.
- **Feature engineering:** compute Greeks, IV, volume/OI ratios, net premium, RVOL.
- **Signal generation:** rules ("unusual call sweep + Vol/OI>3 + short DTE").
- **Filtering/ranking:** reduce the firehose to a short list.
- **Position sizing:** how much to risk (fractional Kelly, fixed-fractional).
- **Risk management:** stops, max exposure, per-name caps.
- **Execution:** manual or broker API.
- **Post-trade analytics:** journal, win rate, expectancy, drift monitoring.

1.3 Options fundamentals from scratch

- **Call option:** the right (not obligation) to BUY 100 shares at the **strike** price before/at **expiry**. You buy calls if you think price rises.
- **Put option:** the right to SELL 100 shares at strike. Buy puts if you think price falls.
- **Premium:** the option's price. One contract = 100 shares, so a \$1.20 premium costs \$120.
- **Intrinsic value:** the in-the-money amount (call: $\max(0, \text{spot} - \text{strike})$). **Extrinsic (time) value:** premium minus intrinsic; decays to zero at expiry.
- **Moneyness:** ITM (in the money), ATM (at the money), OTM (out of the money).
- **Assignment:** the seller's obligation being exercised against them. **American style** (most single-name equity/ETF options) can be exercised any time; **European style** (index options like SPX) only at expiry.
- **Options chain:** the grid of all strikes/expiries with bid/ask, volume, OI, IV, Greeks.
- **Market makers:** provide continuous bid/ask; profit from spread and manage risk by hedging (delta-hedging with shares).
- **Bid/ask/spread:** bid = what buyers offer, ask = what sellers want, spread = the difference; wider spread = higher cost and less liquidity.
- **Open interest (OI):** number of contracts currently open. **Volume:** contracts traded today. Volume creates flow; OI accumulates positioning.

1.4 Options flow / unusual options activity

"Flow" is the time-and-sales stream of individual option prints: contract, size, premium, and inferred aggressor side. Order flow is captured from the consolidated OPRA feed (or, for a free app, a delayed/derived approximation).

- **Sweep:** an order split across multiple exchanges to fill fast — signals urgency. **Golden sweep** is a large, aggressive multi-exchange sweep (often a marketing label for very large-premium sweeps).
- **Block:** a single large negotiated print. **Split:** one order filled by multiple counterparties. **Cross:** a pre-arranged match.
- **Aggressor side / trade-side classification:** did the trade hit the bid (seller-initiated) or lift the ask (buyer-initiated)? This is *inferred*, not given, in delayed data.
 - **Quote rule:** trade above mid = buy, below mid = sell.
 - **Tick rule:** uptick vs previous trade = buy, downtick = sell.
 - **Lee-Ready algorithm:** combine the quote rule with the tick rule for midpoint trades. Full citation: Lee, C.M.C. & Ready, M.J. (1991), "Inferring Trade Direction from Intraday Data," *Journal of Finance* 46(2): 733-746. On accuracy: studies using early-1990s data find Lee-Ready correctly classifies **about 85% of all trades** (Finucane 2000: LR 84.4%, tick test 83.0%; Ellis, Michaely & O'Hara 2000: LR ~81% on Nasdaq; Odders-White 2000 ~77-81%). Accuracy has **degraded with modern high-frequency quoting** — later work (Chakrabarty et al., "Short sales, long sales, and the Lee-Ready algorithm revisited") finds misclassification of up to ~32% of trades in some samples. **Takeaway: treat inferred aggressor side as a useful-but-imperfect ~80% signal, not ground truth.**
- **Repeat flow:** the same contract bought repeatedly across days = accumulation.
- **"Smart money" claims:** a large call buy is NOT proof of informed bullishness — it may be a hedge, one leg of a spread, or dealer-facing. Evidence that raw retail-visible (and delayed) flow reliably predicts returns is weak; treat flow as **attention/positioning evidence**, not a crystal ball.

1.5 Option-buying signals specifically

- **Volume/OI ratio** > 1 (esp. >3): today's volume exceeds existing open interest → likely NEW positioning. The reference site uses **Volume > 3 × OI (with Vol ≥ 200)** as its "Unusual Activity" rule.
- **Large premium trades** (e.g., >\$1M notional): size filter for institutional-scale interest.
- **OTM call/put buying:** cheap leverage bets on direction.
- **Short-dated vs LEAPS:** short DTE = speculative/gamma; LEAPS (long-dated) = longer-term conviction.
- **Put/call ratio:** volume-based sentiment; <0.7 bullish, >1.3 bearish by common convention.
- **Call/put premium imbalance & net premium flow:** dollars into calls vs puts.
- **Delta-weighted / gamma-weighted flow:** weight volume by directional exposure.
- **Dark-pool prints + options flow:** confluence of off-exchange equity blocks and options positioning.
- **IV spikes / skew shifts / term-structure changes:** anticipation of a move.
- **Earnings-related flow & multi-day accumulation:** catalysts and persistence.

1.6 Signal analysis & validation

- **Backtesting:** simulate the rule on history. **Forward testing / paper trading:** run live without money.
- **Look-ahead bias:** using data you wouldn't have had in real time (e.g., same-day close to trade the open). Fatal and common.
- **Survivorship bias:** testing only on stocks that still exist inflates results.

- **Overfitting / data-snooping:** trying many variants and keeping the "best" fits noise.
- **Walk-forward analysis:** roll the train/test window forward through time.
- **Cross-validation for time series:** standard k-fold leaks; use **purged k-fold with embargo** (López de Prado, *Advances in Financial Machine Learning*) to prevent leakage across adjacent samples. His **Combinatorial Purged Cross-Validation (CPCV)** goes further, generating many train/test combinations while respecting chronological order.
- **Deflated Sharpe Ratio (DSR):** corrects the Sharpe ratio for multiple-testing selection bias and non-normal returns; tells you how much "good Sharpe" survives reality. Precise citation: **Bailey, D.H. & López de Prado, M. (2014), "The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting and Non-Normality," *Journal of Portfolio Management* 40(5): 94-107 (SSRN 2460551).** Companion concept: **Minimum Track Record Length** — how long a track record must be before a Sharpe is statistically credible.

1.7 Performance metrics (full explanations in Part 3)

Win rate, expectancy, profit factor, Sharpe, Sortino, Calmar, max drawdown, CAGR, volatility, beta, alpha, information ratio, payoff ratio, Kelly & fractional Kelly, R-multiples.

1.8 Honest limitations

- Free data is **delayed ~15 minutes** because real-time OPRA licensing is expensive.
- Flow signals are **noisy**; many bullish-looking call buys are hedges.
- **Dealer hedging** mechanically distorts prices (see GEX in Part 3).
- A large call buy may be a hedge against a short stock position, or one leg of a spread.
- Realistic expectation: this is an **edge-hunting research tool**, not a money printer.

PART 2 — SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

2.1 Purpose

Specify a locally-run, free-to-use application that ingests US equity/ETF options data, computes Greeks/IV/flow metrics, runs a configurable scanner, backtests signals, and presents dashboards inspired by optionflowthailand.com and comparable platforms.

2.2 Scope

Single-user desktop/local deployment. No cloud dependency required. Free data sources prioritized; paid feeds optional.

2.3 Definitions/Acronyms

OI (open interest), IV (implied volatility), DTE (days to expiration), GEX (gamma exposure), OPRA (Options Price Reporting Authority — the consolidated US options data feed), RVOL (relative volume), ATM/ITM/OTM (moneyness), NBBO (national best bid/offer) — see Glossary (Appendix A).

2.4 Overall description & the reference product

optionflowthailand.com (built by team "20Invest") is a Thai-language US-market analysis SaaS. From direct inspection of the live site, its feature set includes:

- A near-real-time **Options Order Flow** feed (rows: ticker / strike / type / premium, e.g., "NVDA \$140 CALL \$1.2M").
- **Options Radar** (scan the whole market by Volume / CALL-heavy / PUT-heavy / Unusual).
- Per-symbol drill-down: **OI walls** (call wall = resistance, put wall = support), **Max Pain**, **P/C ratio** with an auto "Radar reads it for you" plain-language summary.
- **ODTE** (zero-days-to-expiration) scanner, **Dark Pool** volume (T+1), **Market Tide / Sector Flow heatmap**, **Global money-flow map** (15 markets).
- Support/resistance (pivot points + EMA zones), earnings calendar, 24h news, watchlists (cap 30), a **trade journal** (win-rate / P&L), **portfolio tracker**, insider (SEC Form 4) & 13F institutional tracking, an AI assistant ("Mr.Option"), a Monte-Carlo-style price simulator, and a **Thai capital-gains/options tax calculator**.
- Tiered pricing: Free (฿0), Pro "FlowPro" (฿199/mo), Max "DeepOption" (฿399/mo).

Critically, its own FAQ confirms the free-data reality this manual stresses: its options data (OI, Volume, IV, price) is pulled from CBOE and is **delayed ~15 minutes**, and it explicitly states that real-time per-second data requires an expensive market-data license, so "nearly all analysis platforms use delayed data at this level as standard." Its published formulas match industry conventions: **P/C Ratio = Put Volume ÷ Call Volume** (<0.70 bullish, >1.30 bearish); **Unusual = Volume > 3× OI and Vol ≥ 200**; **Max Pain = strike minimizing total remaining option value** (Option Pain theory); **OI walls** = heaviest-OI strikes used as support/resistance. This is an excellent design benchmark for a free/local clone.

Comparable platforms (feature patterns worth replicating):

- **Unusual Whales** — flagship real-time flow scanner across all US exchanges, GEX analysis, dark-pool data, customizable dashboards, user-configurable alerts (e.g., call volume > 5× daily average), Congress-trade tracker, heatmaps, REST/WebSocket/MCP API; free tier plus paid ("Buffet"/Premium) tiers. Note it is a *data-presentation* layer — it does not generate buy/sell signals.
- **OptionStrat** — flow feed + options profit calculator + live P&L/trade-performance tracking; explains sweeps/splits well.
- **Cheddar Flow / FlowAlgo / BlackBoxStocks** — real-time alerts and unusual-activity feeds.
- **Market Chameleon & Barchart** — unusual-activity screeners, IV rank, earnings tools.
- **QuantData, InsiderFinance** — flow + dark pool dashboards.

Common denominators worth replicating locally: (1) a **flow feed** with sweep/block tags, (2) an **unusual-activity scanner**, (3) **per-symbol chain analytics** (OI walls, max pain, GEX), (4) **alerts**, (5) **watchlists/heatmaps**, (6) a **journal/P&L tracker**.

2.5 Stakeholders & personas

- **Primary user (Pure):** engineer/researcher; wants a local research tool and learning platform.
- **Secondary:** other retail traders if later shared — this **triggers data-redistribution/licensing concerns** (see §2.16).

2.6 Operating environment

Cross-platform (Windows/macOS/Linux), Python 3.11+, local database, optional Docker.

2.7 Functional Requirements (numbered & testable)

- **FR-001 Data ingestion & scheduling:** the system shall fetch option chains and quotes for watchlist symbols on a configurable schedule (e.g., every 15 min during US market hours).
- **FR-002 Watchlist management:** add/remove symbols; cap enforced (e.g., 50).
- **FR-003 Options-chain retrieval & storage:** persist chain snapshots with timestamp.
- **FR-004 Historical archiving:** append daily EOD snapshots to a columnar store.
- **FR-005 Greeks computation:** compute delta/gamma/theta/vega/rho per contract.
- **FR-006 IV computation:** solve implied volatility from mid price.
- **FR-007 Scanner engine:** evaluate configurable rules (Vol/OI, premium, DTE, P/C) and output ranked results.
- **FR-008 Alerting:** push notifications (Telegram/Discord/desktop) when rules fire.
- **FR-009 Backtesting module:** replay signals over history with fees/slippage.
- **FR-010 Paper-trading/journal:** record entries/exits, reasons, P&L.
- **FR-011 Charting & dashboard UI:** price + OI walls + GEX + flow feed.
- **FR-012 Watchlist/heatmap views:** sector flow heatmap.
- **FR-013 Flow feed view:** row-by-row prints with sweep/block/aggressor tags.
- **FR-014 Export:** CSV/Parquet export of any table.
- **FR-015 Configuration management:** YAML/TOML config; hot-reload.
- **FR-016 Local user settings:** theme, refresh interval, alert channels.
- **FR-017 Max pain & GEX:** compute per-symbol max pain and gamma exposure profile.
- **FR-018 Corporate-action handling:** adjust for splits/dividends/symbol changes.

2.8 Non-Functional Requirements

- **NFR-001 Performance:** a watchlist scan of 50 symbols shall complete within one refresh interval; UI queries return in <2 s.
- **NFR-002 Storage footprint:** EOD-only mode shall stay under a few GB/year (see §4.3 math).
- **NFR-003 Reliability:** ingestion retries with exponential backoff; no crash on provider outage.
- **NFR-004 Usability:** every displayed number has an info tooltip (like the reference site's "i" buttons).
- **NFR-005 Portability:** runs on Windows/macOS/Linux.
- **NFR-006 Security:** API keys stored in OS keyring or `.env` (never committed).
- **NFR-007 Maintainability:** modular; typed; tested.
- **NFR-008 Extensibility:** plugin interface for new data providers and signals.
- **NFR-009 Offline behavior:** serve last cached snapshot when offline.
- **NFR-010 Licensing/legal:** respect provider ToS; personal-use only; "not investment advice."

2.9 RECOMMENDED TECHNOLOGY STACK

Core language: Python 3.11+. Justification: dominant quant/data ecosystem, all needed libraries exist, gentle learning curve for an engineer, cross-platform. (Rust/C++ only where hot loops demand it — Numba covers most needs. Note `schwab-py` requires Python ≥ 3.10 .)

Layer	Recommendation	Alternatives	Why
Data frames	polars (heavy) + pandas (glue)	numpy, numba	polars is fast & memory-efficient; pandas for compatibility
Options/quant	py_vollib (IV/Greeks), QuantLib-Python (pricing)	mibian, custom BSM	py_vollib is fast & accurate; QuantLib for advanced/American pricing
Backtesting	vectorbt (research) + backtrader or NautilusTrader (event-driven realism)	zipline-reloaded, PyBroker, backtesting.py	vectorbt for fast parameter sweeps; event-driven for realistic fills
Storage	DuckDB + Parquet files (local-first)	SQLite, TimescaleDB, ClickHouse	DuckDB is columnar OLAP, 10-100× faster than SQLite for analytics, reads Parquet directly, single-file, zero-server
Scheduling	APScheduler	Prefect, cron	in-process, cross-platform, no server
API layer	FastAPI	Flask	async, typed, auto-docs
Frontend	Streamlit (MVP) → optionally React+FastAPI later	Dash, Panel, Electron/Tauri	Streamlit gets a local dashboard running in an afternoon, free, Python-only
Charting	Plotly + TradingView Lightweight Charts	ECharts, lightweight- charts-python	Plotly integrates with Streamlit; Lightweight Charts for candles
Packaging	uv (env/deps) + Docker Compose (optional)	poetry, PyInstaller	uv is fast, reproducible lockfiles
Notifications	Telegram bot + desktop notifications	Discord webhook, email	free, reliable, mobile-friendly

Backtesting design note (2026 practitioner consensus): the standard workflow is a **two-framework stack** — use **vectorbt** to triage thousands of parameter combinations at speed (a 5-year momentum backtest on 500 stocks runs in <1 s vs ~14 s for Backtrader), then re-validate survivors in an **event-driven engine** (NautilusTrader for the most realistic fills and research-to-production parity, or Backtrader for broker integration) before paper trading. **Avoid starting new projects on Backtrader** (effectively frozen) except to reuse existing code; **zipline-reloaded** remains best only for cross-sectional equity factor research via its Pipeline API (painful install).

Storage recommendation (local-first): use **DuckDB as the query engine over Parquet files**, with a small SQLite table for mutable state (settings, journal). DuckDB uses columnar storage and vectorized execution and queries Parquet directly with no import step; Parquet files run ~60-70% smaller than the equivalent SQLite database for financial time series. Mind DuckDB's **single-writer limitation** — fine for a single-user local app.

2.10 FREE AND PAID DATA SOURCES (2026 reality)

The hard truth: true real-time OPRA-level options flow is expensive and effectively never free. Real-time / 15-min-delayed US market data is licensed by exchanges, FINRA, and the SEC, and exchanges charge fees that only paid plans cover. A realistic free/low-cost approximation uses **EOD chain snapshots, volume/OI deltas, delayed quotes, and snapshot-diff-based inferred flow** — exactly what the reference site does with 15-min delayed CBOE data.

Provider	Options chains?	Greeks/IV?	Trade-level flow?	Free tier (2026)	Paid entry	Suitability
yfinance / Yahoo	Yes (delayed, unofficial)	Partial	No	Free, no key, but fragile/unofficial, rate-limited/blocked	—	MVP prototyping only; scrapes Yahoo endpoints, breaks often (broke after Yahoo's Feb 2025 redesign)
Tradier	Yes (chains; Greeks/IV via ORATS)	Yes	Time & sales	Sandbox token (delayed)	\$10/mo data, or free w/ brokerage	STRONG free-ish option: chains + Greeks + T&S
Polygon.io → Massive.com	Yes (paid)	Yes	Yes (tick, paid)	Free: 5 calls/min, 15-min delayed, no options on free	Options Starter ~\$29/mo (15-min delayed); Advanced \$199/mo (tick)	Good when you can pay. Rebranded to Massive.com on Oct 30, 2025, 4 PM ET; <code>api.polygon.io</code> still supported in parallel with <code>api.massive.com</code>
Alpha Vantage	EOD options (free-ish), realtime options (top tiers)	EOD Greeks (premium)	No	25 req/day, 5/min	\$49.99/mo (75/min, EOD options); realtime options on \$199.99+	OK for EOD; sourced from OPRA
Finnhub	Limited	Partial	No	60 calls/min (most generous free)	Paid varies	Good for quotes/news; thin on options
Alpaca	Yes (chains, Greeks, IV)	Yes	Trades/quotes	Free: 15-min delayed REST, real-time IEX, 200 req/min	Algo Trader Plus \$99/mo (full SIP + OPRA options, 10k req/min)	STRONG: brokerage + options data + paper trading
Interactive Brokers	Yes	Yes	Yes (w/ subscription)	Free w/ funded account (data subs	Market-data add-ons	Best if you trade with IBKR

Provider	Options chains?	Greeks/IV?	Trade-level flow?	Free tier (2026)	Paid entry	Suitability
(ib_insync / ib_async)				extra)		
Schwab Trader API (schwab-py)	Yes (options, orders)	Yes	Limited	Free w/ Schwab account (app approval needed)	—	TD Ameritrade API disabled after market close May 10, 2024 ; migrate to schwab-py / Schwab Trader API
ThetaData	Yes (tick)	1st/2nd/3rd-order Greeks	Yes (every OPRA trade paired w/ NBBO)	Free: 30 days EOD	Value \$40/mo, Standard \$80/mo (tick), Pro \$160/mo	Best cheap historical options tick data
Databento (OPRA.PILLAR)	Yes	Compute yourself	Yes (full OPRA, all 17 exchanges)	\$125 free credits (Databento)	Standard \$199/mo OPRA; (Databento) usage-based historical (Databento) still available	Institutional-grade; pay-as-you-go historical; history from Mar 2023 (+10 yrs added back to Apr 2013)
ORATS	Yes	Yes	Aggregates	14-day trial	~\$99/mo+	Best consistent historical Greeks/IV for backtests (also powers Tradier Greeks)
CBOE DataShop / LiveVol	Yes	Yes	Yes	No	Varies (expensive)	Authoritative source
Intrinio	Yes	Yes	Yes	Limited demo	~\$1,000/mo+	Institutional budget only
OptionsDX	Historical EOD/intraday (free)	Pre-calc Greeks/IV	Intraday snapshots	Free historical back to ~2010 (SPX/VIX/SPY etc.)	Paid bulk	GREAT free historical for backtesting

Provider	Options chains?	Greeks/IV?	Trade-level flow?	Free tier (2026)	Paid entry	Suitability
Dolthub options datasets	Historical EOD	Some	No	Free (<u>post-no-preference/options</u> repo, 2019→, ~2,098 symbols)	—	Free bulk historical via <u>dolt clone</u>
MarketData.app	Yes	Yes	No	Free token; 15+ yrs history via Sheets add-on	Cheap paid	Handy for spreadsheets/quick pulls
Nasdaq Data Link (Quandl)	Some	—	No	Limited	Varies	Mostly non-options now
IEX Cloud	—	—	—	SHUT DOWN 31 Aug 2024	—	Do not use. Announced retirement 31 May 2024 (was <2% of IEX revenue, loss-making since inception); assets acquired by former execs Tim Baker & Pedro Aguayo and relaunched as Blue Sky Data
Twelve Data / EODHD / Marketstack	Limited options	Some	No	Small free tiers	Cheap paid	Mostly equities; weak options

Recommended free stack: Tradier sandbox and/or Alpaca free (chains + Greeks + delayed quotes) for live-ish snapshots; **OptionsDX + Dolthub** for free historical backtesting; **yfinance only** as a fallback prototype. **Upgrade path:** ThetaData (\$40–80/mo) for tick history, then Databento (\$199/mo) for full OPRA when serious.

2.11 System architecture



2.12 Database schema (suggested)

- **symbols**(symbol PK, name, type[stock/etf], sector, active bool)
- **option_contracts**(contract_id PK, underlying FK, expiry date, strike numeric, right char[C/P], multiplier int, style)
- **chain_snapshots**(id, contract_id FK, ts timestamp, bid, ask, last, mid, volume, open_interest, iv, delta, gamma, theta, vega, rho) — *partition by date*
- **trades_flow**(id, contract_id FK, ts, price, size, premium, exchange, aggressor char[B/S/?], type[sweep/block/split], is_unusual bool)
- **signals**(id, ts, symbol, rule_name, score, payload JSON)
- **alerts**(id, signal_id FK, channel, sent_ts, status)
- **backtest_runs**(id, strategy, params JSON, start, end, sharpe, cagr, max_dd, profit_factor, created_ts)
- **positions**(id, symbol, contract_id, qty, entry_price, entry_ts, exit_price, exit_ts, pnl)
- **journal**(id, position_id FK, reason_entry, reason_exit, tags, notes)

Indexing: index chain_snapshots on (contract_id, ts) and (underlying, ts); partition Parquet by trade date.

Retention: keep raw EOD snapshots indefinitely (small); prune intraday snapshots after N days.

2.13 API design, config, project structure

REST sketch: `GET /flow?symbol=&min_premium=`, `GET /scan?rule=`, `GET /chain/{symbol}`, `GET /gex/{symbol}`, `GET /maxpain/{symbol}`, `POST /backtest`, `GET /journal`. Config (TOML): providers, api-key references, refresh intervals, scanner thresholds, alert channels. Project layout:

`src/{ingest,storage,greeks,scanner,backtest,api,ui}/`, `config/`, `tests/`, `data/parquet/`, `pyproject.toml`.

2.14 Roadmap (phased MVP)

- **Phase 1:** ingest EOD chains (Tradier/Alpaca/yfinance) → DuckDB; compute Greeks/IV; basic Streamlit chain viewer.
- **Phase 2:** scanner (Vol/OI, premium, P/C) + flow feed + alerts.
- **Phase 3:** GEX/max-pain, sector heatmap, historical archiving, backtester.
- **Phase 4+:** paper trading/journal, broker API, plugin providers, packaging.

2.15 Testing strategy

Unit (Greeks/IV numerics), integration (provider adapters), data-quality tests (no negative prices, monotonic OI dates), **golden-file tests for Greeks** against known BSM values, backtest reproducibility (fixed seed, versioned data).

2.16 Risk register & legal

Provider ToS/scraping risk (yfinance), data-redistribution restrictions (personal use only — both Polygon/Massive and similar vendors draw a sharp line between personal/research use and serving data to your own users), "not investment advice" disclaimers, and market-data license limits if the app is ever shared publicly.

PART 3 — VALUE READING GUIDE

For each metric: definition → formula → worked example → normal range → high/low meaning → misreadings → combine-with.

3.1 Price/market data

- **Last / Bid / Ask / Mid / Spread / Spread%**. $\text{Mid} = (\text{bid} + \text{ask}) / 2$. $\text{Spread\%} = (\text{ask} - \text{bid}) / \text{mid}$. Example: bid 1.00, ask 1.10 → mid 1.05, spread 0.10, spread% ≈ 9.5%. Good: spread% < ~2% (liquid). Bad: >10% (illiquid; you lose a lot to the spread).
- **Volume / Average volume / RVOL**. $\text{RVOL} = \text{today's volume} \div \text{average volume}$. Example: 5M vs 1M avg → RVOL 5.0 (very active). >2 = notable interest.
- **VWAP**. Volume-weighted average price; intraday fair-value benchmark.
- **Market cap** = shares × price. **Float** = tradable shares. **Short interest** = shares sold short; **days to cover** = short interest ÷ avg daily volume. High days-to-cover (>5) = squeeze potential.

3.2 Options-specific

- **Strike, DTE** (days to expiration), **Premium**, **Notional** = premium $\times$ 100 $\times$ contracts (dollar size). **Bid-ask spread on options** (wide is common on illiquid strikes).
- **Open interest, Volume, Vol/OI ratio**. Vol/OI > 1 $\rightarrow$ new positioning; >3 with Vol $\geq$ 200 $\rightarrow$ "unusual" (reference-site rule). **OI change day-over-day** confirms whether new positions stuck.

3.3 The Greeks (intuitive + mathematical)

- **Delta (Δ)**: price change of option per \$1 move in underlying; also a rough probability of expiring ITM. ATM call $\approx$ 0.50. Example: Δ 0.40, stock +\$1 $\rightarrow$ option +\$0.40.
- **Gamma (Γ)**: how fast delta changes; largest ATM and near expiry.
- **Theta (Θ)**: daily time decay; negative for buyers. Example: Θ -0.05 $\rightarrow$ lose \$5/contract/day, all else equal.
- **Vega (ν)**: sensitivity to a 1 vol-point change in IV. Example: vega 0.10, IV +1% $\rightarrow$ +\$0.10.
- **Rho (ρ)**: sensitivity to interest rates (minor for short DTE).
- **Second-order: Vanna** (delta's sensitivity to IV), **Charm** (delta decay over time), **Volga** (vega's sensitivity to IV). These drive dealer hedging drift (see GEX).
- **Delta-adjusted volume** = volume $\times$ delta: converts option activity into equivalent share exposure.

3.4 Volatility

- **Historical/realized vol**: standard deviation of past returns, annualized.
- **Implied volatility (IV)**: the volatility the market prices into options.
- **IV Rank** = (current IV - 52w low) $\div$ (52w high - 52w low). **IV Percentile** = % of days IV was below current. Example: IV 30%, 52w range 20-60% $\rightarrow$ IV Rank 33%. High IV rank (>50) = options relatively expensive (favor selling); low = cheap (favor buying).
- **IV crush**: IV collapses after earnings, hammering long option holders even if direction was right.
- **Smile/skew**: OTM puts usually pricier (crash insurance). **Term structure**: IV across expiries.
- **VIX** = 30-day expected S&P vol. **25-delta risk reversal** = a skew measure.
- **Expected move** $\approx$ ATM straddle price $\approx$ spot $\times$ IV $\times \sqrt{\text{DTE}/365}$. Example: \$100 stock, IV 30%, 30 DTE $\rightarrow 100 \times 0.30 \times \sqrt{(0.082)} \approx$ **\$8.6 expected move**.

3.5 Flow metrics

- **Net premium** = call premium - put premium (dollars). Positive = bullish tilt.
- **Put/Call ratio** (volume or OI). <0.7 bullish, >1.3 bearish (contrarian at extremes).
- **Bullish vs bearish classification**: calls bought at ask + puts sold at bid = bullish; inverse = bearish (with all the caveats of §1.6).
- **Sweep vs block**: urgency vs negotiated size.
- **GEX (gamma exposure)**: aggregate dealer gamma. **Dollar GEX per 1% move** = $\Gamma \times \text{dealer position} \times 100 \times \text{spot}^2 \times 0.01$. (ExpireWorthless) **Positive GEX** $\rightarrow$ dealers dampen moves (range-bound); **negative GEX** $\rightarrow$ dealers amplify moves (trending/volatile). (Gexmetrix) **Gamma flip** = price where net dealer gamma crosses zero. (ExpireWorthless) **Call/put walls** = strikes of largest positive/negative gamma $\rightarrow$

resistance/support. *Caveat: GEX depends on a modeling assumption about the sign of dealer inventory and describes volatility sensitivity, not direction.*

- **Max pain:** strike where the most option value expires worthless; price often drifts toward it into expiry.

3.6 Pricing models

- **Black-Scholes-Merton (BSM):** prices European options from spot, strike, DTE, risk-free rate, and IV. Plain-language: it discounts the probability-weighted payoff. Worked example: $C = S \cdot N(d1) - K \cdot e^{(-rT)} \cdot N(d2)$. For $S=100$, $K=100$, $r=4\%$, $T=0.25$, $\sigma=30\%$: $d1 \approx 0.142$, $d2 \approx -0.008$, $N(d1) \approx 0.556$, $N(d2) \approx 0.497 \rightarrow C \approx 100 \cdot 0.556 - 99.0 \cdot 0.497 \approx \6.4 .
- **Binomial / CRR:** step tree; handles American early exercise.
- **Why American differ:** the early-exercise right adds value (esp. puts, and calls before dividends).
- **Implied volatility solving:** invert BSM for σ via **Newton-Raphson** (fast, uses vega) or **bisection** (robust). Pitfalls: no solution below intrinsic value; vega $\rightarrow 0$ deep ITM/OTM breaks Newton; clamp inputs and fall back to bisection.

3.7 Statistics & backtest outputs

- **Mean/median/stdev/z-score/percentile/correlation** — standard descriptive stats.
- **Sharpe** = (return – risk-free) $\div$ stdev. Example: 12% return, 3% rf, 10% vol $\rightarrow 0.9$. >1 good, >2 excellent, <0 bad.
- **Sortino** = like Sharpe but only downside deviation. **Calmar** = CAGR $\div$ max drawdown.
- **Max drawdown:** worst peak-to-trough loss. **Expectancy** = (win% $\times$ avg win) – (loss% $\times$ avg loss). **Profit factor** = gross profit $\div$ gross loss (>1.5 good).
- **p-values, sample size:** more trades = more confidence; beware few-sample flukes and apply the **deflated Sharpe** correction for many trials.
- **Telling a fake backtest:** suspiciously smooth equity curve, Sharpe >3 on daily data, few trades, no fees/slippage, in-sample only, overfit parameters.

3.8 Cheat-sheet thresholds

Metric	Good	Neutral	Bad	Caveat
Option spread%	<2%	2-10%	>10%	liquidity-dependent
RVOL	>2	1-2	<1	context matters
Vol/OI	>3	1-3	<1	new-position proxy
IV Rank	>50 (sell)	30-50	<30 (buy)	regime-dependent
P/C ratio	<0.7 bull	0.7-1.3	>1.3 bear	contrarian at extremes
Sharpe	>1	0-1	<0	net of costs
Profit factor	>1.5	1-1.5	<1	sample-size
Max DD	<15%	15-30%	>30%	strategy-dependent
Win rate	context	context	context	meaningless without payoff ratio

3.9 Annotated flow row

Example row: NVDA | \$140 CALL | exp 5DTE | 2,000 contracts | premium \$1.2M | at ASK | SWEEP |
Vol/OI=4.2 | IV 55%.

- **Reasonable bullish read:** an aggressive short-dated OTM call sweep at the ask, new positioning (Vol/OI>3), someone paying up for upside into a catalyst.
- **Reasonable alternative read:** could be a dealer/counterparty hedge, one leg of a call spread (a higher strike sold simultaneously), or a roll — not necessarily a naked directional bet. Always check for offsetting prints and news before acting.

PART 4 — MAINTENANCE GUIDE

4.1 Routine checklists

- **Daily:** verify ingestion ran; row counts sane; no stale timestamps; alerts delivered; disk space OK.
- **Weekly:** review signal hit-rate; reconcile journal; check for provider schema changes; back up DB.
- **Monthly:** dependency updates; re-validate top signals; prune intraday snapshots; rotate logs.
- **Quarterly:** full backtest re-run; API-key rotation; disaster-recovery test; Python/lib upgrade review.

4.2 Data pipeline health

Detect missing/stale data ($\max(ts)$ age check), schema drift (validate columns/types on ingest), API deprecations (pin versions, monitor changelogs). Retry with exponential backoff + jitter; respect rate limits. Handle corporate actions: splits (adjust strikes/prices), dividends (affect pricing), symbol changes, delistings, and **non-standard/adjusted options** (e.g., post-split deliverables) — flag adjusted contracts to avoid mispriced Greeks.

4.3 Database maintenance & storage math

Vacuum/compact DuckDB; partition Parquet by date; archive old snapshots to compressed Parquet; back up nightly (copy Parquet + DuckDB file offsite). **Storage math:** $N \text{ symbols} \times M \text{ contracts/symbol} \times R \text{ rows/day}$. Example: $50 \text{ symbols} \times \sim 200 \text{ contracts} \times 1 \text{ EOD snapshot} = 10,000 \text{ rows/day} \approx 2.5\text{M rows/year}$. At $\sim 100 \text{ bytes/row}$ in Parquet $\approx \sim 250 \text{ MB/year EOD}$ — trivial. Intraday every 15 min ($\sim 26 \text{ snapshots/day}$) multiplies by 26 $\rightarrow \sim 6.5 \text{ GB/year}$ — still manageable; prune after N months.

4.4 Dependency & environment management

Pin versions in `pyproject.toml`; commit `uv.lock`; use a virtualenv; **expect yfinance to break** (test after each Yahoo change — it broke after Yahoo's Feb 2025 redesign) and keep a provider-adapter abstraction so you can swap sources instantly. Test Python minor-version upgrades in a branch (remember schwab-py needs ≥ 3.10).

4.5 Signal/model maintenance

Monitor **alpha decay** (rolling hit-rate); re-validate periodically; drift detection (feature distributions shift); retire a signal when out-of-sample edge disappears; re-fit parameters sparingly (over-refitting = overfitting). Track deflated Sharpe across all trials, not just the winner.

4.6 Logging & observability

Structured JSON logging; log rotation; alert on pipeline failure (Telegram); a simple local metrics page (rows ingested, last run, error counts).

4.7 Troubleshooting runbook

Symptom	Likely cause	Fix
Empty option chain	yfinance/Yahoo change or rate-limit	switch provider adapter; add delay/proxy; use Tradier/Alpaca
Stale timestamps	scheduler not running / market closed	check APScheduler; verify market-hours logic
Greeks look wrong	adjusted/non-standard option or bad IV	flag adjusted contracts; clamp IV solver
DB slow	no partitioning/indexing	partition Parquet by date; add DuckDB indexes
Alerts not sent	expired bot token	rotate token; check network
Backtest irreproducible	unpinned data/seed	version data; fix random seed

4.8 Version control & releases

Git; semantic versioning; changelog discipline; tag releases; keep docs in-repo.

4.9 Disaster recovery

DB corruption → restore last Parquet/DuckDB backup. Provider shutdown (like **IEX Cloud, 31 Aug 2024**) → swap adapter to alternate provider. Machine failure → restore from offsite backup; keep infra-as-code (Docker Compose) for fast rebuild.

4.10 Security maintenance

Rotate API keys quarterly; store secrets in OS keyring/`(.env)` (gitignored); scan dependencies for vulnerabilities (`(pip-audit)`); use least-privilege tokens.

APPENDICES

Appendix A — Glossary

ATM/ITM/OTM (moneyness); Delta/Gamma/Theta/Vega/Rho (option sensitivities); OI (open interest); IV (implied volatility); DTE (days to expiration); GEX (gamma exposure); Max Pain (strike where most option value expires worthless); Sweep/Block (aggressive multi-exchange order / negotiated large print); OPRA (consolidated US options feed); NBBO (national best bid/offer); RVOL (relative volume); VWAP (volume-weighted average price); Sharpe/Sortino/Calmar (risk-adjusted return ratios); Kelly (optimal bet-sizing fraction); R-multiple (profit/loss expressed in units of initial risk). Full definitions appear at first use above.

Appendix B — Learning path (engineer → finance)

- **Books:** Hull, *Options, Futures, and Other Derivatives* (foundational pricing); Natenberg, *Option Volatility and Pricing* (trader intuition); Sinclair, *Volatility Trading*; Ernie Chan, *Quantitative Trading & Algorithmic Trading*; López de Prado, *Advances in Financial Machine Learning* (validation methodology — purged k-fold, CPCV, DSR).
- **Free/online:** QuantConnect Bootcamp, CBOE education, university MOOCs on derivatives.
- **Communities:** r/options, r/algotrading, QuantConnect forums, Wilmott.

Appendix C — Common beginner mistakes

Trading illiquid options (wide spreads), ignoring IV crush around earnings, over-leverage, chasing every "unusual" print without context, overfitting backtests, omitting fees/slippage in tests, look-ahead bias, treating a delayed feed as real-time, and mistaking hedges for directional bets.

Appendix D — Realistic expectations ("will this make me money?")

Honestly: **building the app will make you a far better, more disciplined trader and a strong quant engineer; it will NOT reliably print money from free, delayed flow data.** Most retail flow "signals" are noisy and already partly priced in by the time you see them 15 minutes late. Cboe reported record options volumes for a sixth consecutive year in 2025, driven by a 0DTE explosion and ~6.6× more retail options activity vs pre-2020 — meaning *more data but also more noise*. Treat profit as a hypothesis to be tested rigorously, expect months of iteration, and size risk so a wrong hypothesis doesn't hurt.

Appendix E — Sample code (Python)

```
python
```

```

# 1. Fetch an options chain (yfinance MVP fallback)
import yfinance as yf
tk = yf.Ticker("AAPL")
exp = tk.options[0]
chain = tk.option_chain(exp) # chain.calls, chain.puts

# 2. Black-Scholes Greeks + IV (py_vollib)
from py_vollib.black_scholes import black_scholes
from py_vollib.black_scholes.greeks.analytical import delta, gamma, theta, vega
from py_vollib.black_scholes.implied_volatility import implied_volatility
price = black_scholes('c', 100, 100, 0.25, 0.04, 0.30)
d = delta('c', 100, 100, 0.25, 0.04, 0.30)
iv = implied_volatility(6.4, 100, 100, 0.25, 0.04, 'c')

# 3. Detect unusual volume (reference-site rule: Vol > 3*OI and Vol >= 200)
def is_unusual(row):
    return row.volume >= 200 and row.openInterest > 0 and row.volume > 3 * row.openInterest

# 4. Classify aggressor side (quote rule + tick fallback = Lee-Ready core)
def aggressor(price, bid, ask, prev_price):
    mid = (bid + ask) / 2
    if price > mid: return 'B'
    if price < mid: return 'S'
    if prev_price is None: return '?'
    return 'B' if price > prev_price else 'S' if price < prev_price else '?'

# 5. Simple vectorized backtest sketch (vectorbt)
import vectorbt as vbt
close = vbt.YFData.download("SPY").get("Close")
fast, slow = close.vbt.rolling_mean(10), close.vbt.rolling_mean(30)
entries, exits = fast.vbt.crossed_above(slow), fast.vbt.crossed_below(slow)
pf = vbt.Portfolio.from_signals(close, entries, exits, fees=0.001, slippage=0.001)
print(pf.stats())

# 6. Streamlit + FastAPI skeleton
# app.py (Streamlit): st.dataframe(scan_results); st.plotly_chart(gex_fig)
# api.py (FastAPI): @app.get("/scan") def scan(...): return results

```

Appendix F — Legal & regulatory notes (Thailand-based person trading US options)

Not legal, tax, or investment advice. Verify with a professional and with current official sources — rules changed materially in 2026.

-  **FINRA Pattern Day Trader (PDT) rule — REPEALED as of 4 June 2026.** For ~25 years, a "pattern day trader" (4+ day trades in 5 business days, >6% of total trades, in a **margin** account) had to keep **\$25,000 minimum equity** under FINRA Rule 4210. The SEC approved FINRA's amendments on **14 April 2026** (Exchange Act Release No. 105226); FINRA announced them in **Regulatory Notice**

26-10 (20 April 2026), effective **4 June 2026**, which "replace in their entirety the outdated day trading margin requirements, including the day trade count requirements for designating a customer a 'pattern day trader' and the \$25,000 pattern day trader minimum equity requirement." A new real-time "**intraday margin**" framework replaces it (18-month phase-in ending 20 Oct 2027); the ordinary **\$2,000** margin-account minimum still applies. The old rule was account-type-based (margin accounts), never residency-based, so it applied to Thai residents using US brokers too — but this is now moot. *(Any pre-2026 source still describing the \$25k PDT rule as active is outdated.)*

- **Broker access for Thai residents: Interactive Brokers** (top-rated international broker for Thailand) and **tastytrade** (Thailand is on its official eligible-country list; individual cash/margin/joint accounts; W-8BEN required) both explicitly serve Thai residents for US options. **Webull, Firstrade** (international, individual accounts, no SSN needed — verify Thailand inclusion at application), **Zacks Trade, Saxo, TradeStation, MEXEM** also appear among options-capable brokers serving Thailand. **Charles Schwab International's** acceptance of Thai residents was not confirmed — verify directly. Firstrade does **not** accept US citizens/residents living abroad.
- **Thai outward-investment rules (Bank of Thailand):** the BOT's Exchange Control page permits a resident to invest directly in foreign securities up to **USD 5 million per person per calendar year** without an onshore intermediary (up from USD 200,000 in 2019; the Bangkok Post reported a planned raise toward USD 10 million — verify the current figure with BOT/your bank). No limit applies when investing *through* an onshore Thai intermediary. For a typical retail options trader this cap is not binding.
- **US tax for non-resident aliens (NRAs): capital gains — including from options trading — are generally NOT subject to US tax** for an NRA present in the US fewer than 183 days in the tax year (IRC §871; IRS guidance); no US withholding applies to those gains. Exception: if present ≥ 183 days, US-source net capital gains are taxed at a flat 30% (IRC §871(a)(2)). **US-source dividends** paid to NRAs face a flat **30% withholding** (IRC §871(a)), reducible by treaty via **Form W-8BEN**.
- **US-Thailand tax treaty (in force 1998):** under **Article 10**, portfolio dividends to a Thai-resident beneficial owner are capped at **15%** (10% only where the owner is a *company* controlling $\geq 10\%$ of voting power). REIT dividends generally do not get the reduced rate unless the holder owns $< 25\%$. Capital-gains article (13) defers to domestic law, but US law already exempts NRA portfolio gains.
- **Thai personal income tax on foreign trading profits — 2024 change: Departmental Instruction Por.161/2566 (issued 15 Sep 2023, effective for income remitted from 1 Jan 2024)** reinterprets Section 41 of the Thai Revenue Code: a Thai **tax resident** (present ≥ 180 days/year) now owes Thai personal income tax on **foreign-sourced income when it is remitted into Thailand**, in the year of remittance, regardless of when earned — ending the old "wait a year" loophole. **Por.162/2566 (20 Nov 2023)** grandfathers income earned before 1 Jan 2024. Tax is triggered by **remittance**, not by realizing the gain offshore; Thailand's Double Taxation Agreements allow crediting foreign tax paid; LTR-visa holders may be exempt. The Revenue Department has signaled possible further moves toward worldwide-income taxation — **confirm current rules with a Thai accountant at filing time.**